DEFENCE · SECURITY & RESILIENCE

Know if the Wi-Fi is safe before you trust it.

NetGuard checks the security posture of every network you join, then tells you in plain language whether it can be trusted.

linux desktop utility · python, zero dependencies · 10 posture checks · notify-send

THE PROBLEM

Public Wi-Fi asks for trust it never earned

1 click

to join an untrusted network

and the same click can hand over credentials to an attacker.

0 signals

that traffic is manipulated

A padlock means the link is encrypted, not that the network is honest.

Most exposed

have the least support

Remote workers, small teams, journalists and travellers.

NetGuard closes that gap at the exact moment of risk: the network change.

WHY THE USUAL ANSWERS FAIL

Existing help arrives too late, or says too little

TOOLS

Enterprise agents

Built for managed fleets. Too heavy for one laptop on hotel Wi-Fi.

ADVICE

"Just use a VPN"

Sound, but it never tells you whether **this** network is hostile, or why.

THE OS

A padlock icon

Shows link encryption only. Says nothing about DNS, the gateway or interception.

The insight: make the network change the trigger, and turn ten checks into one verdict.

IDEA & INNOVATION

Check the network at the moment you join it

01 · DETECT

Watch the network

Streams NetworkManager events for instant reaction, with a state-fingerprint poller as a fallback.

02 · ASSESS

Run 10 posture checks

Encryption, TLS interception, DNS, gateway, evil twin, portal, proxy, exposure, MAC, reachability.

03 · INFORM

One clear verdict

SAFE, CAUTION or UNSAFE, with the reasons and one concrete action, on the desktop.

Proactive by trigger, explainable by design, reproducible without hardware.

HOW IT WORKS

A small, transparent pipeline

Network change
new link, gateway or DNS

→

State collector
nmcli · ip
iw · resolvectl

→

Check engine
10 posture checks
safe, targeted probes

→

Verdict engine
weighted score
hard severity caps

→

Inform
notify-send
text · json · html

probes: DNS lookups · DNS-over-HTTPS comparison · TLS handshakes to well-known hosts · captive-portal request · port probe on **your own gateway only**. No capture, nothing stored.

THE POSTURE CHECKS

Ten checks, mapped to real attacks

wifi_encryption — HIGH
Open, WEP, WPA1 or TKIP links.

tls_interception — CRITICAL
MITM via untrusted or private-CA certificates.

dns_integrity — HIGH
Hijacked answers vs. a trusted DoH resolver.

arp_gateway — HIGH
Spoofed ARP entries and rogue gateways.

evil_twin — HIGH
Same SSID advertised with different security.

captive_portal — MEDIUM
Portals that intercept and redirect traffic.

proxy_injection — MEDIUM
Transparent proxies rewriting plain HTTP.

router_exposure — MEDIUM
Telnet, UPnP and admin ports on the gateway.

mac_randomization — LOW
Is your hardware address randomised?

reachability — INFO
Context: is there egress at all?

Checks degrade to SKIP when a probe cannot run, so a blocked network never produces a false alarm.

VERDICT & SCORING

From ten results to one decision



WEIGHTED

Severity penalties

Failures subtract from 100 by severity: critical 45, high 25, medium 14, low 7. Warnings cost a fraction.

CAPPED

Can't average out

Any critical failure caps the score at 25. Any high failure caps it at 55. Severe findings stay severe.

EXPLAINED

Reasons and action

A plain verdict, the ordered problems, and one recommended action per finding.

`example: open SSID + TLS interception + DNS hijack + evil twin → score 0/100 → UNSAFE`

USER EXPERIENCE

Two notifications, one decision

NETGUARD · NOW

UNSAFE — GrandHotel_Guest

score 0/100 · 4 failed, 2 warnings
TLS intercepted (untrusted certificate)
DNS answers rewritten
Advice: leave this network, do not enter credentials.

NETGUARD · NOW

SAFE — HomeWiFi-5G

score 100/100 · all posture checks passed.

- A **start notification** fires the instant the network changes.
- The **verdict notification** carries the level, score, top reasons and one action. Unsafe never auto-expires.
- Everything is also written to a report: **text, JSON or shareable HTML**.

DEMONSTRATION

Reproducible scenarios, no special hardware

SCENARIO	STORY	KEY FINDINGS	VERDICT
home-safe	WPA3 home network	All checks pass	SAFE
wired-office	Wired office link	Wi-Fi checks skipped by design	SAFE
cafe-open	Open hotspot + portal	Open link, captive portal, exposed MAC	CAUTION
airport-captive	Airport Wi-Fi	Open link, portal, UPnP exposed	CAUTION
corp-interception	Managed corporate net	TLS inspection by a trusted private CA	CAUTION
hotel-mitm	Compromised hotel net	Evil twin, DNS hijack, TLS interception	UNSAFE

```
netguard demo --list   ·   netguard demo hotel-mitm --save-html report.html
```

PRACTICAL APPLICABILITY

Built to deploy on a typical machine

- **Zero Python dependencies.** Standard library plus tools a Linux desktop already has.
- **One-command user install.** Runs as a systemd user service, no root, enabled at login.
- **Event-driven and quiet.** Works only when the network changes; unprobed checks are skipped, never guessed.

no capture

No traffic is captured or stored; only link metadata, locally.

no cloud

Nothing leaves the machine. Auditable in under 3,300 lines.

ALIGNMENT

How NetGuard meets the brief

CRITERION	WEIGHT	HOW IT IS DELIVERED
Idea & innovation	30%	Posture checks triggered by the network change itself, not a schedule.
Relation to category	20%	Earlier detection of MITM, DNS and evil-twin attacks; better decisions under pressure.
Practical applicability	20%	Zero-dependency, user-level install, real checks against the live network.
Design	20%	Plain-language verdicts, colour-coded notifications, shareable HTML report.
Completeness & value	10%	Working tool, tests, simulation, documentation and service.

HONESTY & ROADMAP

What it does well, and what comes next

KNOWN LIMITATIONS

- Evil-twin detection is **heuristic**; a perfect clone needs beacon fingerprinting.
- TLS checks trust the system CA store, so an admin-installed proxy CA is a warning.
- Checks reflect the moment of connection.

ROADMAP

- Passive **802.11 beacon and RTT fingerprinting** for stronger evil-twin detection.
- DNS-over-TLS enforcement and per-process connection attribution.
- Tray indicator and a **fleet agent** for small organizations.

DEFENCE · SECURITY & RESILIENCE

Make the safe choice automatic.

NetGuard turns the instant of a network change into an automatic security posture check, so people, not just enterprises, can tell when public Wi-Fi is not safe to use.

`netguard watch` · `python3 -m netguard demo --list` · `./scripts/install.sh`